RAG vs Fine-Tuning

Two ways to give a language model knowledge it doesn't have

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1. The problem

A model such as Llama 3.2 has general knowledge from its training, but it knows **nothing about your organisation's private information** (products, policies, documents), and its knowledge is **frozen** at the end of training. Asked about GadgetHub's return policy, it can only guess.

There are two main ways to fix this: **RAG** and **fine-tuning**.

> **Analogy**
>
> **RAG** is an **open-book exam**: the model is handed the right page of the handbook with each question.
>
> **Fine-tuning** is a **closed-book exam**: the model is trained in advance until it answers from memory.

2. RAG (Retrieval-Augmented Generation)

RAG **looks up relevant information and adds it to the prompt** each time a question is asked. The model itself is not changed.

1. **Index**: split your documents into small chunks and store them for searching, usually in a vector database.
2. **Retrieve**: for each question, find the most relevant chunks.
3. **Augment**: add those chunks to the prompt alongside the question.
4. **Generate**: the model writes its answer using that information.

Example of what the model receives:

```
[system]  Answer using only these sources:
          [Returns policy] Items can be returned within 30 days.
          [Returns policy] Headphones must be returned unopened.
[user]    Can I return my headphones?
```

3. Fine-tuning

Fine-tuning **trains an existing model further on your own examples**, changing its internal weights so the knowledge or behaviour becomes part of the model. Afterwards it answers without needing extra information in the prompt.

1. **Prepare examples**: hundreds to thousands of question-and-answer pairs in the style you want.

2. **Train**: run training on a GPU, usually with an efficient method such as LoRA.
3. **Test and deploy**: check the new model's answers, then use it in place of the original.

Example of one training example:

```
[user]       Can I return an item?
[assistant]  You can return items within 30 days. Headphones
             can only be returned unopened.
```

4. Key differences

	RAG	Fine-tuning
What changes	What is sent to the model (the prompt)	The model itself (its weights)
Best for	Facts and knowledge	Style, tone, format and specialised behaviour
Updating information	Edit the documents; takes effect immediately	Retrain the model; slow and costly
Data needed	Your existing documents	Many prepared example conversations
Set-up cost	Lower	Higher (data preparation, GPU training)
Cost per question	Higher (search step, longer prompts)	Lower (short prompts)
Shows sources	Yes	No
When it doesn't know	Can say so	Tends to guess
Main weakness	Depends on finding the right information	Knowledge goes out of date

5. When to use which

- Use **RAG** when the model needs **facts** it doesn't know, especially if they **change often** or answers must **show their sources**.
- Use **fine-tuning** when you need a **consistent style or behaviour**, or a narrow, repetitive task, that good instructions alone can't achieve.
- Use **both** together when needed: fine-tune for **how** to answer, and use RAG for **what** the facts are.